

Uniform asymptotic solution for electromagnetic reflection and diffraction of an arbitrary Gaussian beam by a smooth surface with an edge

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Abstract. A closed form solution is obtained to describe, in a physically appealing manner, the reflection and diffraction of a general astigmatic Gaussian beam which is incident on an arbitrary smooth, electrically large, slowly varying curved, perfectly conducting screen (or reflector). This closed form solution is obtained via an asymptotic evaluation of the radiation integral for the fields scattered from the reflector, to within the physical optics approximation that remains valid for the present situation. The analysis developed here is particularly well suited for the fast analysis of electrically large reflector antennas by representing the feed illumination by a relatively small set of Gaussian beams launched from the feed plane. Each of these Gaussian beams after being launched undergoes reflection and diffraction at the reflector; the expressions for the reflected and diffracted fields are developed in this paper and utilized by Chou [1996] to compute the radiation pattern of large reflector antennas in a matter of a few seconds as compared to the conventional numerical physical optics integral method which takes hours on the same computer.

1. Introduction

A closed form uniform asymptotic solution is developed in this paper for predicting the reflection and diffraction of an arbitrary Gaussian beam (GB) by a general curved surface containing an edge which is depicted in Figure 1. It is noted that the present uniform solution for reflection and diffraction of an arbitrary GB by an edged surface that is otherwise smooth is seen to be analogous to the ray solution based on the uniform asymptotic theory (UAT) or the uniform geometrical theory of diffraction (UTD) for the reflection and diffraction of an arbitrary “ray optical” field when it

illuminates the same type of geometry [Kouyoumjian and Pathak, 1974; Pathak, 1988; Lee and Deschamps, 1976]. Consequently, the uniform asymptotic theory for GBs developed here retains all the advantages of the UAT or UTD based ray solutions. Furthermore, this uniform solution for GBs remains valid in regions of ray caustics and also in the regions of a confluence of ray caustic and ray shadow boundaries where the UAT and UTD edge diffraction solutions fail. A confluence of ray caustic and ray shadow boundary transition regions occurs, for example, when computing the far-zone radiation patterns of paraboloidal reflector antennas within the region of the main beam and the first few side-lobes. In the latter case, ray methods based on the UAT and UTD fail because the caustics of reflected and diffracted rays coincide with the reflection shadow boundary, and one usually then has to resort to an integral technique

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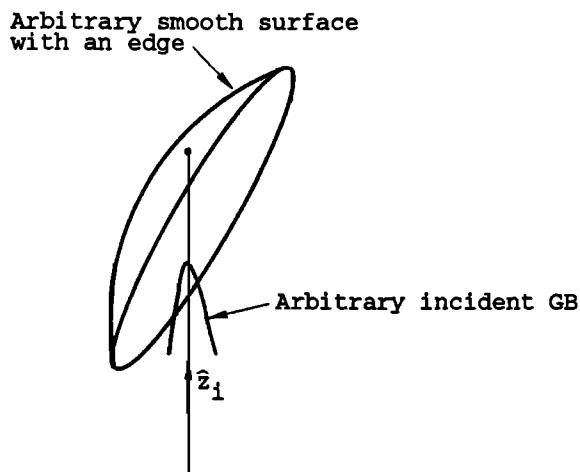


Figure 1. A general astigmatic Gaussian beam (GB) illuminating a smooth reflector surface containing an edge. The incident GB axis is oriented along the direction $\hat{z}_1$ as shown.

such as that based on the physical optics (PO) method, where the integration is performed over currents induced on the reflector surface, or that based on the integration of fields over an appropriate radiating aperture formed by the reflector geometry. However, the latter conventional PO and aperture integral techniques which are employed to calculate the main beam and first few sidelobes of reflector antennas become very costly and inefficient, especially for electrically large antennas, because the integrations in these techniques, in general, have to be performed numerically. On the other hand, the GB based method developed here is extremely fast since it completely avoids the integrations required in the PO and aperture integral techniques and because this GB solution is expressed in closed form; in addition, the GB technique provides a physical insight into the formation of the radiation pattern. Consequently, the present GB method offers a significant advantage over the numerical PO and aperture fields methods. As indicated later, the feed radiation can be expressed as a superposition of the fields of a relatively small set of GBs, each of which is launched from the feed plane; once launched, these GBs

reflect and diffract as described in this paper to form the reflector radiation pattern. The actual GB expansion for the feed radiation and the computation of reflector antenna patterns using the GB technique developed here is the subject of a separate paper (H.T. Chou et al., A novel Gaussian beam method for the rapid analysis of large reflector antennas, submitted to *IEEE Transactions on Antennas and Propagation*, 1996)(hereinafter referred to as submitted manuscript, 1996).

The GB method developed here can also be modified and extended to deal with electrically large lens type antennas and the synthesis of shaped reflectors or feed arrays for parabolic reflectors or the synthesis of aperture distributions in planar phased array applications and to produce contour beams, spot beams, or the desired scanning characteristics. This method may also be applicable to the analysis/design of millimeter wave antennas whose radiation fields can typically be modeled by GBs, as well as to analyze their interaction with most millimeter wave components to predict diffraction losses which can become an important consideration in their design. These latter topics are presently under investigation.

The medium external to the edged structure shown in Figure 1 is assumed to be free space. Although the curved surface is taken here to be a perfect electric conductor (PEC), this analysis can be extended to an impedance surface as well, but the latter is not considered in this paper. It is assumed that the incident GB is not close to grazing on the curved surface. Also, the edged surface which is otherwise smooth is assumed to be electrically large and slowly varying. The incident GB can be arbitrary in that it represents a general astigmatic electromagnetic GB with the plane of incidence that is not necessarily aligned with the principal surface directions at the point of reflection associated with the incident GB axis. Here the plane of incidence is defined by the incident GB axis and the unit normal to the surface at the real point of reflection, and the real point of reflection is simply the intersection of the incident GB axis with

the surface which it illuminates. If the incident GB axis does not intersect the reflecting surface that is truncated by the edge, i.e., if the incident GB axis lies beyond the surface containing the edge, then in this special case the incident GB axis can be viewed as intersecting the analytical extension of the original reflecting surface past the edge. It is further assumed that the polarization of the general astigmatic incident GB is orthogonal to the GB axis; this assumption can be shown to remain valid in the paraxial region of the GB. The latter assumption is usually met in the applications to reflector antennas and other practical problems where GBs can be utilized. It is noted that a rotationally symmetric GB, an elliptical GB, and an astigmatic GB are, respectively, all special cases of the general astigmatic GB. The preceding is a result of the fact that first, an astigmatic GB contains two distinct waists at two distinct locations along the GB axis, and if the locations of these two waists happen to coincide, then the GB is said to be elliptical, whereas if the locations and the sizes of the two waists become identical, then the GB becomes rotationally symmetric. Furthermore, the coordinates or the principal axes along which the exponential amplitude matrix and the phase matrix of the GB become separately diagonal are no longer coincident for the general astigmatic GB, and they rotate relative to each other as a function of the axial beam propagation distance until the far-zone conditions are reached. When these exponential amplitude and phase terms become diagonal along the same real principal axes, then the general astigmatic GB simply reduces to an astigmatic GB.

In order to find the fields reflected and diffracted by a smooth surface containing the edge when it is illuminated by an arbitrary GB, one can conveniently employ the PO approximation for the scattered field. The PO approximation is valid and adequate because the slowly varying surface is assumed to be electrically large and it is well illuminated by the incident GB. While the analysis presented in this paper can be used in the case when the observer lies behind the re-

flector, only the case where the observer lies in the region $z > 0$ of Figure 2, with $\hat{z}_i \cdot \hat{z} < 0$, is primarily considered in this paper. A further discussion of this aspect is given in section 4. The PO integral over the reflecting surface for an incident GB is evaluated here in closed form using a uniform asymptotic procedure. For an incident GB which strikes the reflecting surface sufficiently far from any part of the edge, the resulting edge diffraction effects become negligible in comparison to reflection effects because of the exponentially decaying nature of the incident GB away from its axis. However, when the incident GB strongly illuminates any part of the edge (which can happen if the incident GB axis is directed close to an edge), then the GB undergoes composite reflection and diffraction; that is, the incident GB produces a nonnegligible edge-diffracted field component, whereas the rest of the incident GB energy is transformed into a modified reflected wave. The precise modification of the reflection process due to nonnegligible edge diffraction is described in the uniform asymptotic solution in terms of an error function which automatically provides a proper transition from one of essentially entire reflection when edge diffraction effects are negligible to a modification of reflection when diffraction effects tend to become significant. If the incident GB misses the surface and is also far from the edge, then only diffraction effects remain. The details of this asymptotic development are

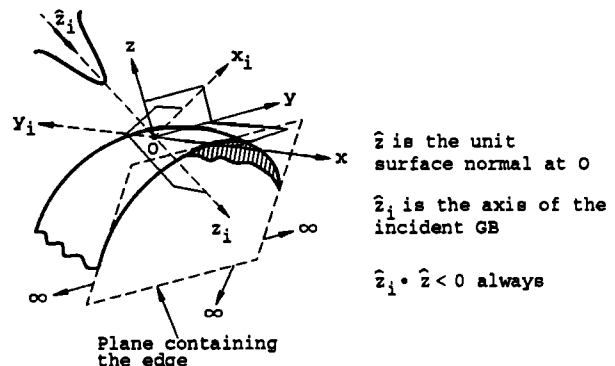


Figure 2. Arbitrary GB incident from an arbitrary direction on the reflecting surface containing a planar edge.

presented later in this paper. As indicated earlier, an engineering application of this asymptotic closed form solution for the reflection and diffraction of an arbitrary incident GB by an edge is described in a separate paper (Chou et al., submitted manuscript, 1996) to analyze a realistic offset parabolic reflector antenna with a given feed excitation. It is shown by Chou et al., (submitted manuscript, 1996) that the feed radiation can be synthesized in terms of a set of relatively few GBs according to some simple rules; each of these GBs is then incident on the reflector from where it simply reflects and diffracts according to the solution obtained here. The utility, efficiency and accuracy of such a GB based reflector analysis are evident from the results shown by Chou et al., (submitted manuscript, 1996) who show that the GB based solution agrees extremely well with a reference numerical PO solution that took several hours of computer time versus only a few seconds for the present GB based solution.

Keller and Streifer [1971] and *Deschamps* [1971] have previously described the GB field as being associated with a bundle of complex rays; furthermore, they also indicated that a GB field in two or three dimensions is equivalent to the paraxial field of a complex line source or a complex point source, respectively. The terms complex line and point sources are meant to indicate that those sources lie in an appropriate complex coordinate space; the latter extension of source location from a real to complex space allows one to employ usual ray tracing procedures but in "complex" space in order to evaluate the scattered field via complex rays when a GB is incident on a large scattering object. Such a complex source point (CSP) approach has been further extended by *Shin and Felsen* [1978], *Hasseltmann and Felsen* [1982], and *Maciel and Felsen* [1990]. Although the CSP method is accurate, it appears to be cumbersome for numerical calculations in practical situations, except in some special instances, because the points of ray reflection and diffraction, etc., have, in general, to be searched numerically in complex space. More importantly, it

has not yet been shown to be possible to use the CSP technique to model a general astigmatic or an arbitrary GB which is considered in this paper. It is noted that a general astigmatic GB is produced even in the simple instance when a rotationally symmetric GB (which, of course, can be generated via the CSP technique) is incident obliquely on any smooth curved reflecting surface; thus an arbitrary GB is often encountered in practice. Some simpler techniques for tracking GBs in the presence of reflecting surfaces which are based on phase matching were employed by *Massey and Siegman* [1969] and by *Burkholder and Pathak* [1991]. The simpler phase-matching technique assumes a priori that the reflected field is also a GB when the incident field is a GB; thus such a technique works well in situations where the incident GB spot size is sufficiently small in comparison with the radii of curvature of the reflecting surface because, otherwise, the resultant reflected field is not a pure GB due to beam distortion caused by surface curvature. Also, the phase-matching technique does not provide an estimate of the edge-diffracted fields if the reflecting surface is truncated as it usually is, in practice. *Suedan and Jull* [1987] employ the complex ray or CSP method [*Shin and Felsen*, 1978; *Hasseltmann and Felsen*, 1982; *Maciel and Felsen*, 1990] in conjunction with the UTD [*Kouyoumjian and Pathak*, 1974; *Pathak*, 1988] to find the far-out sidelobes and back lobes of diffraction by slits in planar screens and by planar as well as parabolic reflectors in two dimensions. Their method [*Suedan and Jull*, 1987, 1991] works well in regions where the reflected GB is absent (except for the planar reflector); however, it fails in the case of the parabolic reflector within regions where the reflected field is present and constitutes the dominant contributor to the scattered field. A fully three-dimensional (3-D) parabolic reflector analysis that utilizes complex ray tracing is described by *Hasseltmann and Felsen* [1982], and it deals with a situation where the incident GB from the feed only weakly excites the edge; it therefore does not include any edge diffraction contribution. As pointed out

earlier, the more conventional numerical evaluation of the PO integral still appears to be the most popular way among engineers to analyze reflector antenna problems; however, as also mentioned previously, the numerical PO integration method is extremely inefficient for large reflector surfaces. More recently, *Anastassiou and Pathak* [1995], and *Zogbi* [1994] have described some preliminary approaches for analyzing the reflection and diffraction of an incident GB by a finite curved reflector based on a closed form asymptotic evaluation of the PO integral in the 2-D and 3-D cases, respectively. In particular, the 2-D results of *Anastassiou and Pathak* [1995] for prediction of the edge diffraction are valid only in the far-zone of the 2-D reflector, and the 3-D solution of *Zogbi* [1994] for edge diffraction is valid only for the case when the plane tangent to the local paraboloid, at the point of reflection on a smooth curved surface, is perpendicular to the plane containing the edge. Consequently, the 3-D solution of *Zogbi* [1994] cannot, for example, treat an offset parabolic reflector antenna. In contrast, the present 3-D solution for the reflection and edge diffraction of an arbitrary incident electromagnetic (EM) GB overcomes the limitations of *Zogbi* [1994] in that it allows for the reflection and diffraction of an arbitrary incident GB even for a planar curved edge when it may not be orthogonal to the tangent plane at the point of reflection; thus the present result obtained here is applicable to offset reflector antennas as well. In addition, the solution is far more compact than the previous one, and it contains fewer approximations than those in the previous method [*Zogbi*, 1994], thereby making the present solution also more accurate. The closed form nature of the present solution also results from the use of a local paraboloid approximation at the real point of reflection of an arbitrary GB which is incident on the actual smooth reflector with an edge; furthermore, this allows one to avoid any complex ray tracing thereby making it highly efficient. For the sake of completeness, it may be remarked that *Heyman and Ianconescu* [1995] have analyzed the diffraction of a pulsed beam by a wedge; while the analysis

of *Heyman and Ianconescu* [1995] is done in the time domain and also provides useful physical insight into the behavior of the scattered wave packet, it is nevertheless restricted to a straight edge formed by a wedge with plane faces. On the other hand, the present analysis is developed for the time harmonic case, but the edge which the incident GB illuminates can now be a general curved edge in a curved reflecting surface.

The format of the paper is as follows. The field of an arbitrary incident GB is described in section 2, together with the details of the PO integral formulation for the field scattered from the reflector. Also, the various coordinate systems required for subsequent analysis are described in section 2. A uniform asymptotic evaluation of the PO integral formulation of section 2 is described in section 3; this asymptotic analysis leads to closed form expressions for the edge-diffracted and edge-reflected field contributions in which the latter includes modifications arising from the presence of edge diffraction. Some numerical results are presented in section 4 to illustrate the accuracy and speed of the GB solution. An $e^{+j\omega t}$ time dependence for the fields is assumed and suppressed in the following analysis. The $e^{+j\omega t}$ above as usual represents an exponential function with an exponent $j\omega t$, in which $j = \sqrt{-1}$, $\omega = \text{angular frequency} = 2\pi f$ (f in Hz), and only here does t refer to time.

2. An Integral Formulation for the Fields Scattered From the Reflecting Surface When it is Illuminated by an Arbitrary GB

Figure 2 illustrates an arbitrary, EM GB which is incident on an arbitrary, smooth PEC surface from an arbitrary direction. Let O be the real point of reflection where the incident GB axis strikes the reflector surface. The arbitrary incident GB has the following form for the magnetic field intensity $\bar{H}^i$ associated with it:

$$\bar{H}^i(\bar{r}_i) = \bar{H}^i(O) \sqrt{\frac{\det[Q^i(z_i)]}{\det[Q^i(0)]}} \times e^{-jk(z_i + \frac{1}{2}[\hat{\epsilon}_i]^T Q^i(z_i) \hat{\epsilon}_i)} ;$$

$$[\xi_i] = \begin{bmatrix} x_i \\ y_i \end{bmatrix}, \quad (1a)$$

$$\bar{H}^i \cdot \hat{z}_i \simeq 0; \quad \bar{E}^i(\bar{r}_i) \sim \hat{z}_i \times Z_0 \bar{H}^i(\bar{r}_i) \quad (1b)$$

where $\bar{r}_i = (x_i, y_i, z_i)$ is the position vector in the incident GB coordinate system and $\bar{H}^i(O)$ is the incident magnetic field at the origin O for which $\bar{r}_i = (0, 0, 0)$. The $\bar{E}^i$ is the electric field associated with $\bar{H}^i$. Also, k is the wavenumber of free space, Z_0 is the free space impedance, and $k = 2\pi/\lambda$ where λ is the free space wavelength. It is noted that the incident GB propagates in the $\hat{z}_i$ direction, where z_i constitutes the axis of the incident GB and the approximations in (1b) are commonly used within the paraxial region where (1a) is valid. The coordinates (x_i, y_i, z_i) which are fixed in the incident GB also have their origin at O as shown in Figure 2. The matrix Q^i in (1) will be defined shortly; $\det[Q^i]$ in (1a) therefore refers to the determinant of this matrix Q^i . It is assumed that the surface at O can be approximated locally by a paraboloid with the same principal radii of curvature as that of the original surface at O , namely,

$$z \simeq -\frac{1}{2} \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} \frac{1}{R_1} & 0 \\ 0 & \frac{1}{R_2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \quad (2)$$

where the (x, y, z) coordinates are fixed so that $\hat{z}$ is normal to the original surface and the local paraboloid at the point O . The R_1 and R_2 are the two principal surface radii of curvatures along the principal surface directions $\hat{x}$ and $\hat{y}$, respectively, at O . Only the case $R_1 \times R_2 > 0$ is considered in the following analysis, but the case $R_1 \times R_2 < 0$ can be treated in exactly the same fashion. One could also model the surface in the vicinity of O more accurately, if desired, by a local second degree surface which is more general than the local paraboloid, namely, by

$$z = ax + by + cx^2 + dy^2 \quad (3)$$

in which a, b, c , and d must be found by matching (3) to the original surface at and near O including the neighboring points on the edge. However, it is found that edge diffraction is only significant when O is near an edge on the actual

surface (or on its mathematical extension past the edge) because the GB decays exponentially transverse to the direction $\hat{z}_i$. The approximation in (2) is thus found, in general, to be adequate for O at and near the edge. For O far from the edge, only the reflection effects remain significant again because of the exponential decay of the GB transverse to $\hat{z}_i$, and (2) is found to be adequate, in general, for this situation as well. It is for these reasons that the local paraboloidal surface approximation of (2) will be used instead of the somewhat more complicated local surface approximation of (3). In some special cases it may be necessary to use a local bicubic approximation to the surface, and the present approach can also be extended to treat this special case as will be demonstrated in a future communication.

The GB field in (1a) and (1b) is similar to a ray optical field except that the curvature matrix Q^i is complex for the GB case while it is real for the ray optical field. The complex phase or curvature matrix Q^i for the incident GB at $\bar{r}_i = (0, 0, 0)$ is assumed to be

$$Q^i(0) = \begin{bmatrix} Q_{11}^i & Q_{12}^i \\ Q_{12}^i & Q_{22}^i \end{bmatrix} \quad (4)$$

and the $Q^i(z_i)$ at z_i follows the propagation rule of the GB given by

$$[Q^i(z_i)]^{-1} = [Q^i(0)]^{-1} + z_i I \quad (5)$$

which is also true for ray optical fields. I above is the 2×2 identity or unit matrix. The eigenvalues $\lambda_{1,2}$ of the complex curvature (or, alternatively, the complex phase) matrix $Q^i(0)$ are given by

$$\begin{aligned} \lambda_{1,2} &= \frac{1}{\rho_{1,2}^i} \\ &= \frac{1}{2} \left[(Q_{11}^i + Q_{22}^i) \pm \sqrt{(Q_{11}^i + Q_{22}^i)^2 - 4(Q_{11}^i Q_{22}^i - Q_{12}^i)^2} \right] \end{aligned} \quad (6)$$

in which $1/\rho_{1,2}^i$ are the complex principal curvatures of the incident GB; it is noted that these become real principal wavefront curvatures for a

ray optical field in real space. It is noted that if Q^i is a diagonal matrix with identical diagonal terms, then the incident GB is rotationally symmetric; on the other hand, if the diagonal terms of Q^i are different with off diagonal terms still zero, then the incident GB becomes elliptical. However, when the cross-diagonal or off-diagonal terms are nonzero in the matrix Q^i ; i.e., when all the elements of Q^i are nonzero, then one obtains a general astigmatic incident GB, which is simply referred to here as an arbitrary incident GB.

The transformation relating the paraboloidal surface coordinates (x, y, z) at O to the incident GB coordinates (x_i, y_i, z_i) , whose origin is also at O for convenience, is given by

$$\begin{bmatrix} x_i \\ y_i \\ z_i \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (7)$$

which can be written compactly as

$$[r_i] \equiv [c_{ij}] [r_j] \quad (8)$$

where the elements of the transformation $(c_{11}, c_{12}, \text{etc.})$ are known once (x, y, z) and (x_i, y_i, z_i) are specified.

The incident GB strongly illuminates the reflecting surface over the region of its spot size around O . The PO integral for the electric $\bar{E}^s$ and magnetic $\bar{H}^s$ fields scattered from the surface are, respectively, given by

$$\bar{E}^s(\bar{r}) \cong \frac{jkZ_0}{4\pi} \int \int_{S_{lit}} \hat{R} \times \bar{R} \times \bar{J}_{eq}(\bar{r}') \frac{e^{-jkR}}{R} dS' \quad (9)$$

$$\bar{H}^s(\bar{r}) \cong -\frac{jk}{4\pi} \int \int_{S_{lit}} \hat{R} \times \bar{J}_{eq}(\bar{r}') \frac{e^{-jkR}}{R} dS' \quad (10)$$

where S_{lit} denotes only the part of the reflecting surface which is directly illuminated by the incident GB and the equivalent current $\bar{J}_{eq}$ in (9) and (10) is given within this PO approximation by

$$\bar{J}_{eq}(\bar{r}') = 2\hat{n} \times \bar{H}^i(\bar{r}'). \quad (11)$$

It is noted that k and Z_0 denote the wavenumber and impedance of free space, respectively.

The R in (9) and (10) is given, as usual, by

$$\bar{R} = \bar{r} - \bar{r}'; \quad R = |\bar{R}| \quad (12)$$

with

$$\bar{r} = x\hat{x} + y\hat{y} + z\hat{z}; \quad \bar{r}' = x'\hat{x} + y'\hat{y} + z'\hat{z} \quad (13)$$

in which $\bar{r}$ represents the coordinates at some observation point off the surface S_{lit} and $\bar{r}'$ represents the coordinates (also in the (x, y, z) system) of any point on S_{lit} . Thus

$$R = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}. \quad (14)$$

It is noted that terms of $O(1/R^2)$ and higher are ignored in (9) and (10); thus (9) and (10) are valid as long as the observer is not extremely close to S_{lit} .

In (11), $\bar{H}^i(\bar{r}')$ denotes the incident magnetic field at the position $\bar{r}'$ on the surface, which is given by (1a). The unit vector $\hat{n}$ is the normal to the surface S_{lit} at and near O and is given from (2) by

$$\hat{n} = \frac{\hat{z} + \hat{x} \frac{x'}{R_1} + \hat{y} \frac{y'}{R_2}}{\sqrt{1 + \left(\frac{x'}{R_1}\right)^2 + \left(\frac{y'}{R_2}\right)^2}}. \quad (15)$$

The integration along the local parabolic surface in (9) and (10) is, in general, difficult, hence it is desirable to map the integration surface from the local parabolic surface to a projected ($z = 0$) surface via

$$dS' = \sqrt{1 + \left(\frac{x'}{R_1}\right)^2 + \left(\frac{y'}{R_2}\right)^2} dx' dy'. \quad (16)$$

Using (11), (15), and (16), the differential current element can now be written as

$$\bar{J}_{eq}(\bar{r}') dS' = 2 \left[\hat{z} + \hat{x} \frac{x'}{R_1} + \hat{y} \frac{y'}{R_2} \right] \times \bar{H}^i(\bar{r}') dx' dy' \quad (17)$$

where it follows from (1a) that the incident mag-

netic field on the surface in the neighborhood of the reflection point is given by

$$\bar{H}^i(\bar{r}') = \bar{H}_m^i(\bar{r}'_i) e^{jkq(\bar{r}'_i)} \quad (18)$$

with

$$\bar{H}_m^i(\bar{r}'_i) = \bar{H}^i(O) \sqrt{\frac{\det[Q^i(z_i)]}{\det[Q^i(0)]}} \quad (19)$$

$$q(\bar{r}'_i) = - \left[z'_i + \frac{1}{2} (Q_{11}^i x_i'^2 + 2Q_{12}^i x'_i y'_i + Q_{22}^i y_i'^2) \right] \quad (20)$$

in which $\bar{r}'_i$ is the value of $\bar{r}_i$ at (x', y', z') ; i.e., $[\bar{r}'_i] = [c_{ij}][r'_j]$ following (7) and (8). Using (17), (18), and (20), the scattered electric and magnetic fields in (9) and (10) can be written, with the understanding that z' on dS' is a function of (x', y') via (2), as

$$\bar{E}^s(\bar{r}) \cong \frac{jkZ_0}{4\pi} \int \int_{S_p} 2 \left[\hat{R} \times \hat{R} \times (\hat{z} + \hat{x} \frac{x'}{R_1} + \hat{y} \frac{y'}{R_2}) \times \bar{H}_m^i(\bar{r}'_i) \right] \frac{e^{-jk[R-q(x',y')]} dx' dy'}{R} \quad (21)$$

$$\bar{H}^s(\bar{r}) \cong -\frac{jk}{4\pi} \int \int_{S_p} 2 \left[\hat{R} \times (\hat{z} + \hat{x} \frac{x'}{R_1} + \hat{y} \frac{y'}{R_2}) \times \bar{H}_m^i(\bar{r}'_i) \right] \frac{e^{-jk[R-q(x',y')]} dx' dy'}{R} \quad (22)$$

where S_p denotes the projected area of the local paraboloid onto the $z = 0$ plane and $q(\bar{r}') = q(x', y')$ via (2).

It is observed that (21) and (22), in fact, have the same general form given by

$$\bar{P}^s(\bar{r}) = \int \int_{S_p} \bar{F}(x', y') e^{jkf(x', y')} dx' dy' \quad (23)$$

where $f(x', y')$ denotes the phase term of the integrand and is defined by

$$f(x', y') = -R + q(x', y'). \quad (24)$$

In (23), $\bar{P}^s(\bar{r})$ represents either $\bar{E}^s(\bar{r})$ or $\bar{H}^s(\bar{r})$ according to the definition of $\bar{F}(x', y')$, namely,

$$\bar{F}(x', y') = \begin{cases} \frac{jkZ_0}{2\pi R} \left[\hat{R} \times \hat{R} \times (\hat{z} + \hat{x} \frac{x'}{R_1} + \hat{y} \frac{y'}{R_2}) \times \bar{H}_m^i(\bar{r}'_i) \right]; & \bar{E}^s(\bar{r}) \\ \frac{-jk}{2\pi R} \left[\hat{R} \times (\hat{z} + \hat{x} \frac{x'}{R_1} + \hat{y} \frac{y'}{R_2}) \times \bar{H}_m^i(\bar{r}'_i) \right]; & \bar{H}^s(\bar{r}). \end{cases} \quad (25)$$

In the following analysis one assumes that the observer can come within the Fresnel region of the illuminated (bright) spot that the incident GB makes on the reflecting surface. This Fresnel zone approximation is therefore valid even when the observation point is relatively quite close to the original, entire reflector surface. In the Fresnel region, R in the phase term $f(x', y')$ of (24) can be expanded in a Taylor series about the GB reflection point O in which one retains up to quadratic terms in x' and y' . Thus

$$R \approx r - x' \frac{x}{r} - y' \frac{y}{r} + \frac{x'^2}{2r} \left[\left(1 - \frac{x^2}{r^2} \right) + \frac{z}{R_1} \right] + \frac{y'^2}{2r} \left[\left(1 - \frac{y^2}{r^2} \right) + \frac{z}{R_2} \right] - \frac{x'y'}{r} \left(\frac{xy}{r^2} \right), \quad (26)$$

in phase. Note that the amplitude term $1/R$ in $\bar{F}(x', y')$ is not approximated at all.

The approximation in (25) allows one to express the phase term $f(x', y')$ of (24) as [Zogbi, 1994]

$$f(x', y') = -r - a_0 x'^2 + b_0 x' - a_1 y'^2 + b_1 y' - 2cx'y' \quad (27)$$

where the coefficients a_0, a_1, b_0, b_1 and c are defined by Zogbi [1994] as

$$a_0 = \frac{1}{2} \left[\frac{1}{R_1} \left(\frac{z}{r} - c_{33} \right) + \frac{1}{r} \left(1 - \frac{x^2}{r^2} \right) + c_{11}^2 Q_{11}^i + c_{21}^2 Q_{22}^i + 2c_{11}c_{21}Q_{12}^i \right] \quad (28)$$

$$b_0 = \left(\frac{x}{r} - c_{31} \right) \quad (29)$$

$$a_1 = \frac{1}{2} \left[\frac{1}{R_2} \left(\frac{z}{r} - c_{33} \right) + \frac{1}{r} \left(1 - \frac{y^2}{r^2} \right) + \right]$$

$$c_{12}^2 Q_{11}^i + c_{22}^2 Q_{22}^i + 2c_{12}c_{22} Q_{12}^i \quad (30)$$

$$b_1 = \left(\frac{y}{r} - c_{32} \right) \quad (31)$$

$$c = \frac{1}{2} \left[c_{11}c_{12}Q_{11}^i + c_{21}c_{22}Q_{22}^i + (c_{11}c_{22} + c_{21}c_{12})Q_{12}^i - \frac{xy}{r^3} \right]. \quad (32)$$

It is observed that $f(x', y')$ in (27) can also be written in a more compact form by

$$f(x', y') = f(x_s, y_s) - a_0(x' - x_s)^2 - a_1(y' - y_s)^2 - 2c(x' - x_s)(y' - y_s) \quad (33)$$

where x_s and y_s are defined by

$$x_s = \frac{a_1 b_0 - b_1 c}{2(a_0 a_1 - c^2)} \quad (34)$$

$$y_s = \frac{a_0 b_1 - b_0 c}{2(a_0 a_1 - c^2)} \quad (35)$$

and

$$f(x_s, y_s) = -r + \frac{a_1 b_0^2 + a_0 b_1^2 - 2b_1 b_0 c}{4(a_0 a_1 - c^2)}. \quad (36)$$

It can be verified that (x_s, y_s) constitutes the stationary phase point in two dimensions (x', y') and that $f(x_s, y_s)$ is the value of the complex exponential term in the integrand of (23) when it is evaluated at this stationary point. The relevance of this $f(x_s, y_s)$ will become more apparent when dealing with the asymptotic approximation of (23) for large k ; such an asymptotic treatment of (23) leads to essentially a closed form result for the double integral in (23) as discussed in the next section.

In order to simplify the evaluation of (23), let a coordinate system (x_2, y_2, z_2) be introduced at O in which the coordinate direction $(\hat{x}_2)$ is chosen to be perpendicular to the plane containing the edge as in Figure 3. The location of that plane containing the edge is given by $x_2 = x_b$; furthermore, the direction of $\hat{x}_2$ is chosen such that $x_b \geq 0$ always. The remaining rectangu-

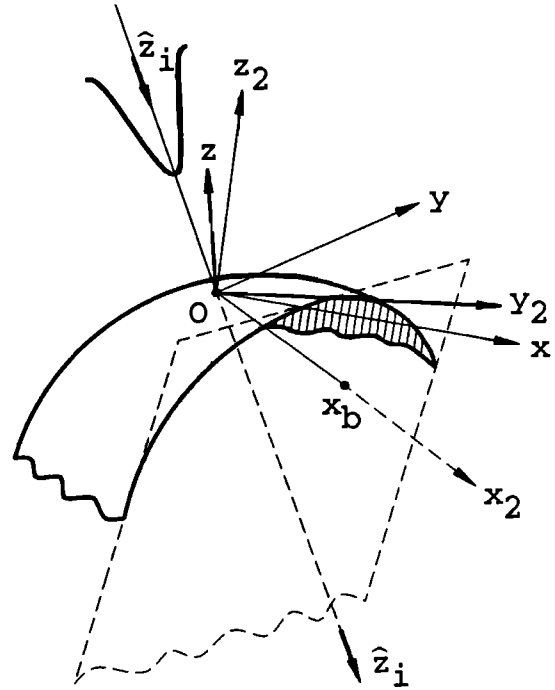


Figure 3. The coordinate system (x_2, y_2, z_2) at O with $\hat{x}_2 \perp$ plane containing the edge. The plane containing the edge is defined by $x_2 = x_b$.

lar coordinates y_2 and z_2 may be chosen relatively arbitrarily. The coordinates (x, y, z) and (x_2, y_2, z_2) at O are related by

$$x_2 = b_{11}x + b_{12}y + b_{13}z \quad (37a)$$

$$y_2 = b_{21}x + b_{22}y + b_{23}z \quad (37b)$$

$$z_2 = b_{31}x + b_{32}y + b_{33}z. \quad (37c)$$

Let $(x = x_e, y = y_e, \text{ and } z = z_e)$ represent the coordinates of the edge. It is also noted that the edge lies in the plane $x_2 = x_b$; thus, from (37a) it is clear that at (x_e, y_e, z_e) , x_2 satisfies

$$x_2 = x_b = b_{11}x_e + b_{12}y_e + b_{13}z_e \quad (38)$$

in which z_e can be approximated near O by (2) as

$$z_e \cong -\frac{1}{2} \left(\frac{x_e^2}{R_1} + \frac{y_e^2}{R_2} \right) \quad (39)$$

so that (38) becomes, via (39), the following

$$x_b = b_{11}x_e - \frac{b_{13}}{2R_1}x_e^2 + b_{12}y_e - \frac{b_{13}}{2R_2}y_e^2. \quad (40)$$

As expected, (40), which is the projection of the local edge in the $z = 0$ plane, is locally a part of an ellipse; this can be seen explicitly by writing (40) as

$$\frac{(x_e - x_o)^2}{A^2} + \frac{(y_e - y_o)^2}{B^2} = d^2 \quad (41)$$

where

$$x_o = \frac{b_{11}R_1}{b_{13}} \quad (42a)$$

$$y_o = \frac{b_{12}R_2}{b_{13}} \quad (42b)$$

$$A^2 = \left| -\frac{2R_1}{b_{13}} \right| \quad (43a)$$

$$B^2 = \left| -\frac{2R_2}{b_{13}} \right| \quad (43b)$$

$$d^2 = \left| x_b - \frac{b_{11}^2 R_1}{2b_{13}} - \frac{b_{12}^2 R_2}{2b_{13}} \right|. \quad (43c)$$

It is assumed in (41) that $(R_1 \times R_2) > 0$ so that the local surface is a paraboloid at 0. If $(R_1 \times R_2) < 0$, then (40) is a hyperbola and can be treated in the same way as above; however, the hyperbola case will not be discussed here. In (41), the point $(x = x_o, y = y_o)$ in the $z = 0$ plane constitutes the center of the ellipse. It is useful to introduce the mapping

$$x_e = x_o + Ad \cos \psi_1 \quad (44a)$$

$$y_e = y_o + Bd \sin \psi_1 \quad (44b)$$

in which A , B , and d are chosen to remain positive. One could also, following (44a) and (44b), introduce

$$x = x_o + A\rho_1 \cos \psi_1 \quad (45a)$$

$$y = y_o + B\rho_1 \sin \psi_1 \quad (45b)$$

so that when $x = x_e$ and $y = y_e$, then $\rho_1 = d$. Thus, in the (ρ_1, ψ_1) coordinates on the $z = 0$ plane, the projected edge is defined as a function of a single variable ψ_1 since $\rho_1 = d$ on the

projected edge. Let the origin O in the $z = 0$ plane be located at $(\rho_1 = \rho_0; \psi_1 = \phi_0)$ in the (ρ, ϕ) system, where

$$\rho_0 = \sqrt{\frac{x_o^2}{A^2} + \frac{y_o^2}{B^2}} \quad (46a)$$

$$\cos \phi_0 = -\frac{x_o}{A\rho_0} \quad (46b)$$

$$\sin \phi_0 = -\frac{y_o}{B\rho_0} \quad (46c)$$

with $\rho_0 > 0$. If $\rho_0 = 0$, then one requires $\phi_0 = 0$; otherwise, ϕ_0 becomes undefined. It is convenient to introduce ϕ_1 , where

$$\psi_1 = \phi_1 + \phi_0. \quad (47)$$

Thus incorporating (47) into (45) yields

$$x = x_o + [A_1 \cos \phi_1 + A_2 \sin \phi_1] \rho_1 \quad (48a)$$

$$y = y_o + [B_1 \cos \phi_1 + B_2 \sin \phi_1] \rho_1 \quad (48b)$$

where

$$A_1 = A \cos \phi_0 \quad (49a)$$

$$A_2 = -A \sin \phi_0 \quad (49b)$$

$$B_1 = B \sin \phi_0 \quad (49c)$$

$$B_2 = B \cos \phi_0. \quad (49d)$$

The variable ϕ_1 in (47) is introduced because the significant contribution to the integral in (23) comes from the neighborhood of (ρ_0, ϕ_0) , which is the region where the incident GB most strongly illuminates the original surface, and this, in turn, indicates that ψ_1 is close to ϕ_0 or ϕ_1 is small within that region of interest (typically, $\phi_1 \ll \pi$). The relations between the (x, y) and (ρ, ϕ) coordinates are given in Appendix A.

It is thus possible to transform (23) expressed in the (x, y) coordinates into the following integral in the $(\rho_1; \phi_1)$ coordinates of integration:

$$\bar{P}^s(\bar{r}) = \int_0^d \int_{-\pi}^{\pi} \bar{G}(\rho_1, \phi_1) e^{jkg(\rho_1, \phi_1)} d\phi_1 d\rho_1 \quad (50)$$

where

$$\bar{G}(\rho_1, \phi_1) \equiv \bar{F}(x', y') |J| \quad (51)$$

$$g(\rho_1, \phi_1) \equiv f(x', y') \quad (52)$$

in which the Jacobian J of the transformation is given explicitly via Appendix A as

$$\begin{aligned} J &= \frac{\partial x'}{\partial \phi_1} \frac{\partial y'}{\partial \rho_1} - \frac{\partial x'}{\partial \rho_1} \frac{\partial y'}{\partial \phi_1} \\ &= -AB\rho_1. \end{aligned} \quad (53)$$

It is noted that the limits on ρ_1 in (50) are valid for the illumination of the surface S_{lit} on a finite reflector which is of interest here. This analysis can also be directly extended to a different case in which a surface complementary to S_{lit} is illuminated by a GB; however, this complementary surface that would be of infinite extent will project an area that lies outside of the ellipse of Figure 4 so that the limits on ρ_1 would

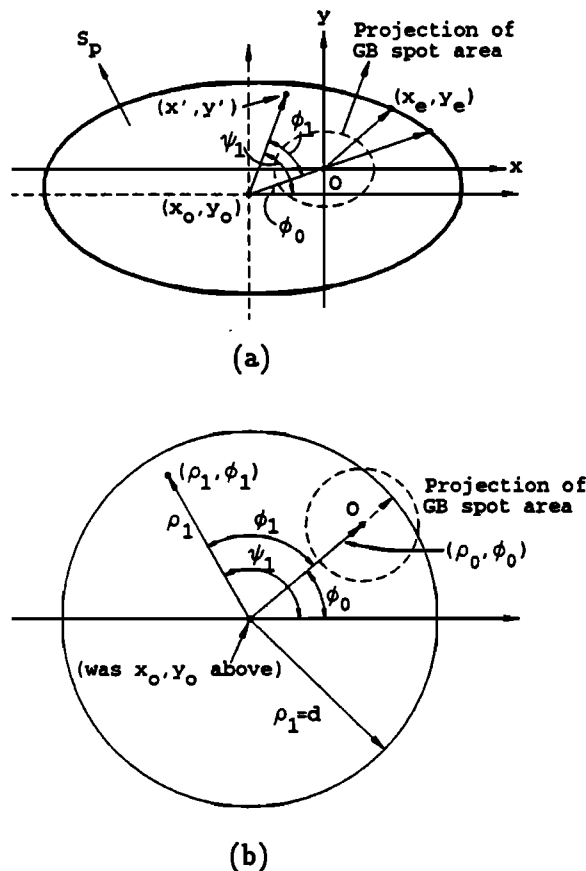


Figure 4. (a) The projected area of the local parabolic surface S on the $z = 0$ plane and (b) its mapping to the (ρ_1, ψ_1) coordinate system.

be changed appropriately in that instance. The PO based integral in (50) with the phase approximation in (33) will next be evaluated asymptotically in closed form in section 3.

3. Closed Form Asymptotic Evaluation of the Integral Formulation for the Scattered Fields

In order to evaluate the double integral in (50) in closed form, one may reduce (50) asymptotically for large k . Initially, one may consider the case when the real point O , which is at the center of the projection of the incident beam on the $z = 0$ plane, is in the neighborhood of some part of the projected reflector edge on the same $z = 0$ plane as shown in Figure 4. The projection of the incident GB on the $z = 0$ plane decays away from O , and for an electrically large reflector this projected field is negligible at (x_o, y_o) in Figure 4a which corresponds to $\rho_1 = 0$ in Figure 4b. Consequently, for this case under consideration, (50) may be written symbolically as

$$\bar{P}^s(\bar{r}) = \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \bar{G}(\rho_1, \phi_1) e^{jkg(\rho_1, \phi_1)} d\rho_1 d\phi_1. \quad (54)$$

For large k the dominant contribution to (54) must come from the neighborhood of O (at $\rho_1 = \rho_0, \phi_1 = 0$) because e^{jkg} decays away from $(\rho_1 = \rho_0; \phi_1 = 0)$. The ϕ_1 integration will be evaluated first. Let

$$\bar{P}^s(\bar{r}) = \int^d \bar{P}_\phi(\rho_1) d\rho_1 \quad (55)$$

where

$$\bar{P}_\phi(\rho_1) = \int_{-\pi}^{\pi} \bar{G}_\phi(\phi_1) e^{jk_g \phi(\phi_1)} d\phi_1 \quad (56a)$$

$$\bar{G}_\phi(\phi_1) = \bar{G}(\rho_1, \phi_1) \quad (56b)$$

$$g_\phi(\phi_1) = g(\rho_1, \phi_1), \quad (56c)$$

with ρ_1 fixed. It is noted that $\bar{G}_\phi(\phi_1)$ is not singular in $(-\pi < \phi_1 < \pi)$ and it is slowly varying in the neighborhood of the saddle point at $\phi_1 = \phi_{1s}$ where

$$g'_\phi(\phi_{1s}) = 0 \quad (57)$$

and the prime on g'_ϕ denotes the operation $\partial/\partial\phi_1$. Evaluating (56) asymptotically via the saddle point method yields, after retaining only the leading term [Felsen and Marcuvitz, 1966]:

$$\bar{P}_\phi(\rho_1) \approx \bar{G}_\phi(\phi_{1s}) \frac{\sqrt{-2\pi}}{\sqrt{jk g''_\phi(\phi_{1s})}} e^{jk g_\phi(\phi_{1s})} \quad (58)$$

where the branch of each square root in the numerator and the denominator of (58) denoted here by Z (with $Z = -2\pi$ in the numerator or $Z = jk g''_\phi(\phi_{1s})$ for the denominator) that is chosen separately to satisfy $\text{Im}[\sqrt{Z}] \geq 0$. Incorporating (57) into (56) yields

$$\bar{P}^s(\bar{r}) = \int_d^d \bar{G}_\rho(\rho_1) e^{jk g_\rho(\rho_1)} d\rho_1 \quad (59)$$

where

$$\bar{G}_\rho(\rho_1) \equiv \bar{G}_\phi(\phi_{1s}) \frac{\sqrt{-2\pi}}{\sqrt{jk g''_\phi(\phi_{1s})}} \quad (60)$$

$$g_\rho(\rho_1) \equiv g_\phi(\phi_{1s}) = g(\rho_1, \phi_{1s}). \quad (61)$$

It is important to note that $\phi_{1s} = \phi_{1s}(\rho_1)$. Evaluating (59) asymptotically by the saddle point method, taking into account the end point $\rho_1 = d$ [Felsen and Marcuvitz, 1966], yields, after retaining the leading terms,

$$\bar{P}^s(\bar{r}) \sim \bar{P}_r(\bar{r}) \cdot T(S_a) + \bar{P}_d(\bar{r}). \quad (62)$$

The saddle point in the integrand of (59) resides at $\rho_1 = \rho_{1s}$ where

$$g'_\rho(\rho_{1s}) = 0 \quad (63)$$

and the prime on g'_ρ denotes the operation $\partial/\partial\rho_1$. The various terms in (62) are given by

$$\bar{P}_r(\bar{r}) \equiv \bar{G}_\rho(\rho_{1s}) \sqrt{\frac{-2\pi}{jk g''_\rho(\rho_{1s})}} e^{jk g_\rho(\rho_{1s})} \quad (64)$$

$$T(S_a) \equiv 1 - \frac{1}{2} [1 - \text{erf}(S_a)] + \frac{e^{-S_a^2}}{2\sqrt{\pi} S_a} \quad (65)$$

$$\bar{P}_d(\bar{r}) \equiv \bar{G}_\rho(d) \frac{1}{jk g'_\rho(d)} e^{jk g_\rho(d)}. \quad (66)$$

The $\text{erf}(S_a)$ in (65) denotes the error function whose argument S_a is given by

$$S_a \equiv \sqrt{jk [g_\rho(\rho_{1s}) - g_\rho(d)]}. \quad (67)$$

It can be seen from (33) that the interior stationary phase point of the double integral in (23) (and hence (50)) is given by $x' = x_s$ and $y' = y_s$, where the coordinates of the stationary phase point (x_s, y_s) are explicitly defined in (34) and (35), respectively. The values of ρ_{1s} and ϕ_{1s} are the location of this same stationary point in the (ρ_1, ϕ_1) coordinates. It is thus observed that if (64) is produced by a stationary point that lies interior to the projected area S_p , then (64) represents the reflected field contribution alone when an arbitrary GB is incident on the reflecting surface as if the edge were absent. This reflected field component in (64) can be simplified and expressed back in terms of the original variables F and f in (23) as

$$\bar{P}_r(\bar{r}) = -\pi \bar{F}(x_s, y_s) \frac{1}{\sqrt{-jka_1}} \frac{1}{\sqrt{-\frac{jk(a_0 a_1 - c^2)}{a_1}}} e^{jk f(x_s, y_s)}. \quad (68)$$

The subscript r on $\bar{P}_r$ is used specifically to denote that it represents only the reflected field contribution. It is important to note that one must choose the imaginary part of each square root in (68) to be positive. One also notes that (68) can be obtained directly via the nonuniform, double-stationary phase asymptotic evaluation of (23) (which essentially assumes that there are no edges on projected surface S_p of integration). Furthermore, it can be shown that a paraxial approximation to (68) yields a reflected general astigmatic GB as might be expected; this paraxial approximation is given by Zogbi [1994]. In most instances one needs to add correction terms to the paraxial approximation of (68) as indicated by Zogbi [1994]. On the other hand, $\bar{P}_r$ in (68) does not contain the

usual paraxial approximation along the direction of reflection of the axial ray of the incident GB which strikes the reflector surface, and hence it is expressed more compactly and more accurately than by Zogbi [1994].

The transition function T in (65) which results from a uniform asymptotic evaluation of the ρ_1 integral in (59) automatically modifies the reflected field $\bar{P}_r$ when the incident GB illuminates the reflecting surface close to an edge, thereby correcting the reflected field due to effects of edge diffraction, which become important in this situation since now only part of the incident GB is, in a sense, reflected. It is also convenient to simplify (67) and express the argument S_a of the error function in (65) in terms of the original $f(x', y')$ of (23). Indeed, it is clear from (52) and (61) that

$$S_a = \sqrt{jk[f(x_s, y_s) - f(x_{se}, y_{se})]} \quad (69)$$

where (x_s, y_s) are defined in (34) and (35), and x_{se} and y_{se} are given by (44a) and (44b) when they are specialized to the value of the saddle point ϕ_{1s} at $\rho_1 = d$; in particular,

$$\phi_{1s}(\rho_1)|_{\rho_1=d} \equiv \phi_{1se}. \quad (70)$$

The ϕ_{1se} can be found in closed form as described in Appendix B. The branch of the square root in (69) is chosen so that

$$\begin{aligned} \text{sign}(\text{Re}[S_a]) = \\ \text{sign} \left\{ \text{Re} \left[\sqrt{-\frac{jk}{2} g''_\rho(\rho_{1s}) \times (d - \rho_{1s})} \right] \right\} \end{aligned} \quad (71)$$

with

$$\text{Re} \left[\sqrt{-\frac{jk}{2} g''_\rho(\rho_{1s})} \right] > 0. \quad (72)$$

The right-hand side of (71) is valid for small S_a , which is true within the transition region where the reflection effects are only partially present owing to modification by diffraction which begins to be nonnegligible. For large S_a , one is

outside the transition region where $T \rightarrow 1$ if $\text{Re}[S_a] > 0$ and where $T \rightarrow 0$ if $\text{Re}[S_a] < 0$. Thus the following results are obtained: When $S_a \rightarrow 0$,

$$T \approx 1/2 + \frac{1}{2\sqrt{\pi}S_a}$$

and

$$\begin{aligned} \bar{P}^s(\bar{r}) &\rightarrow \frac{1}{2}\bar{P}_r(\bar{r}) + \Delta \\ \left(\Delta = \left[\frac{\bar{P}_r}{2\sqrt{\pi}S_a} + \bar{P}_d(\bar{r}) \right] \rightarrow 0 \right). \end{aligned}$$

When S_a is large and $\text{Re}(S_a) > 0$, $T \rightarrow 1$ and $\bar{P}^s(\bar{r}) \rightarrow \bar{P}_r(\bar{r}) + \bar{P}_d(\bar{r})$. When S_a is large and $\text{Re}(S_a) < 0$, $T \rightarrow 0$ and $\bar{P}^s(\bar{r}) \rightarrow \bar{P}_d(\bar{r})$, which corresponds to the situation where (x_s, y_s) lies outside of the projected area S_p .

The edge-diffracted field $\bar{P}_d$ produced by the interaction of the incident GB with the edge of the reflecting surface remains to be discussed. The result for $\bar{P}_d(\bar{r})$ in (66) can be simplified in terms of F of (23) as

$$\begin{aligned} \bar{P}_d(\bar{r}) = \\ \bar{F}(x_{se}, y_{se}) \frac{ABd}{jk g'_\rho(d)} \frac{j\sqrt{2\pi} e^{jk f(x_{se}, y_{se})}}{\sqrt{jk g''_\phi(\phi_{1se})}}. \end{aligned} \quad (73)$$

It is shown in Appendix B that ϕ_{1se} of (70) can be obtained in closed form. The $g'_\rho(d)$ and $g''_\phi(\phi_{1se})$ are also shown explicitly in Appendix B.

It is noted that the result in (62) is not directly applicable to the case when GB is incident close to the axis of a rotationally symmetric reflector such that the spot size of the incident GB near the reflector is large compared to the reflector size. The latter case must be treated separately, but the details of such a separate treatment are avoided here since this case is not of interest in the present work. Only incident GB spot sizes which are sufficiently small compared to the reflector size are of interest in this study.

4. Summary and Numerical Results

The electromagnetic fields scattered by a reflecting surface containing an edge when it is illuminated by an arbitrary incident GB are given formally by the PO radiation integrals in (21) and (22). A general form of the PO radiation integral is given by (23) with $\bar{F}$ as in (25) for the scattered electric field $\bar{E}^s$ and for the scattered magnetic field $\bar{H}^s$, respectively. An asymptotic evaluation of $\bar{P}^s$ in (23) gives a closed form result for the scattered field which is summarized in (62) as

$$\bar{P}^s(\bar{r}) \sim \bar{P}_r(\bar{r}) \cdot T(S_a) + P_d(\bar{r}). \quad (74)$$

The $\bar{P}_r(\bar{r})$ is an interior (double) stationary point contribution corresponding to the "reflected" field component in the region where the observer is located such that $z > 0$ with $\hat{z}_i \cdot \hat{z} < 0$. The region for $z > 0$ is of primary interest in this paper. Observation points in the region $z < 0$ yields a result for $\bar{P}^s(\bar{r})$ behind the reflector which for large $S_a > 0$ (and hence $T \rightarrow 1$) leads to $\bar{P}^s(\bar{r}) \sim -\bar{P}_i(\bar{r}) + P_d(\bar{r})$, where the stationary phase contribution $\bar{P}_r(\bar{r})$ reduces automatically within the paraxial approximation to the negative of the incident GB represented above by $-\bar{P}_i$ in which

$$\bar{P}_i = \left\{ \begin{array}{l} \bar{E}_i \\ \bar{H}_i \end{array} \right\}; \quad \left\{ \begin{array}{l} \text{electric} \\ \text{magnetic} \end{array} \right\} \quad (75)$$

field case as expected.

Some numerical results are presented below to illustrate the accuracy of the above GB based method. Additional numerical results for the radiation from practical, electrically large offset reflector antennas are given by Chou et al., (submitted manuscript, 1996), who also provide some details of a GB expansion for the feed radiation. Specifically, the numerical results for the reflection and diffraction of a single incident GB by a curved reflecting surface with an edge are shown in Figures 5–7, which are based on the closed form GB solution compared with the corresponding independent solution based on the conventional numerical PO integration. It is

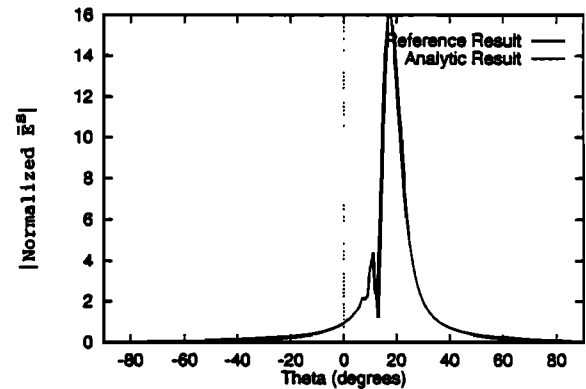
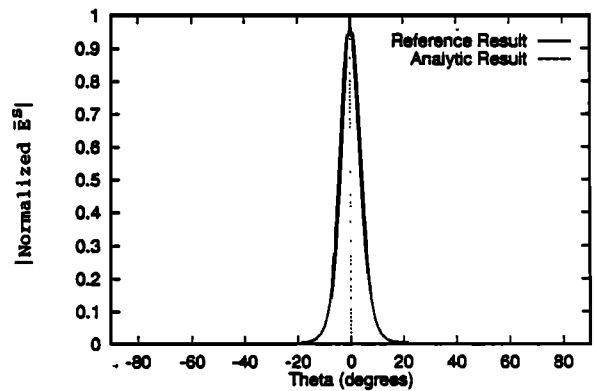
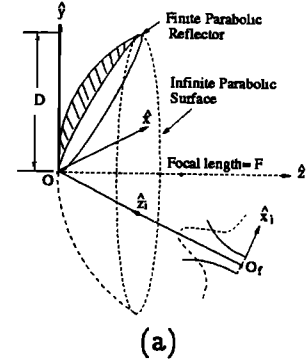


Figure 5. (a) Geometry of three-dimensional (3-D) Gaussian beam scattering from a finite curved surface. (b) Axial beam reflection point is on the edge, $\phi = 0$ degrees plane. (c) Beam waist is located at $(0, -20\lambda, 50\lambda)$, $\phi = 90$ degrees plane. Focal length, $F = 36\lambda$ and projected diameter, $D = 40\lambda$.

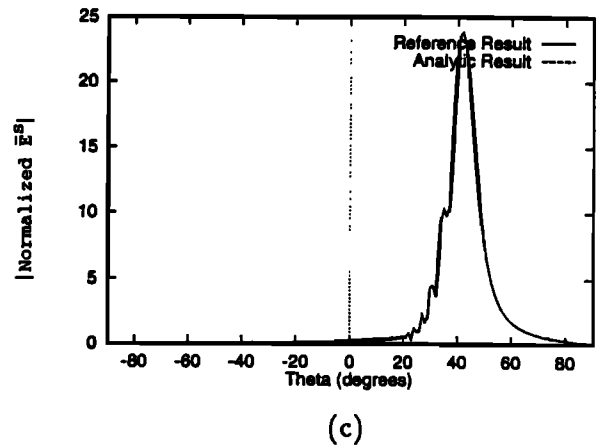
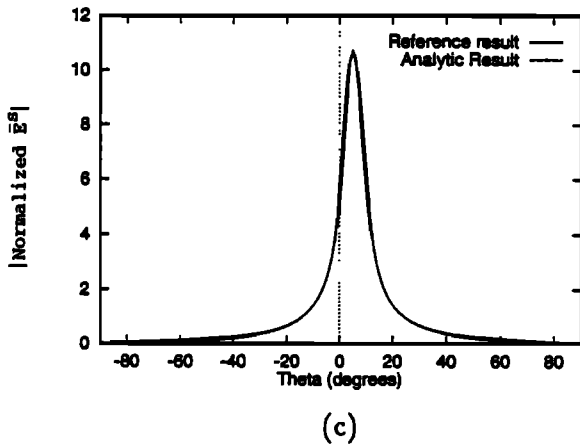
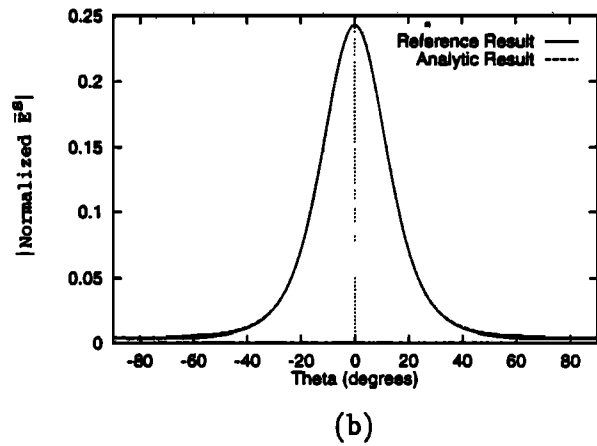
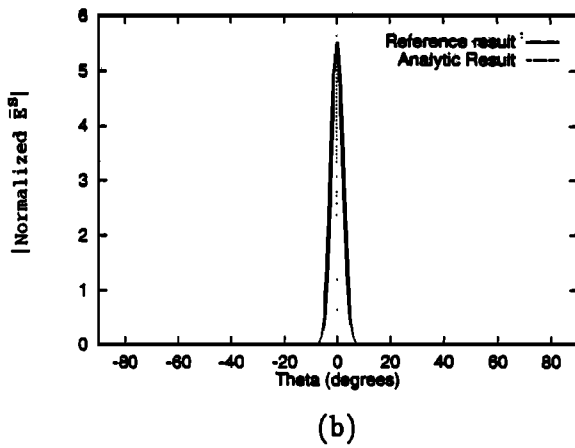
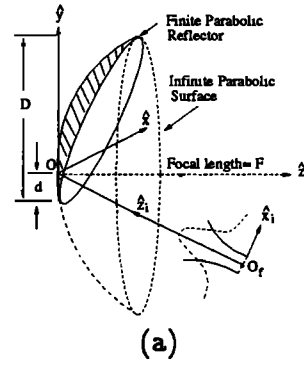
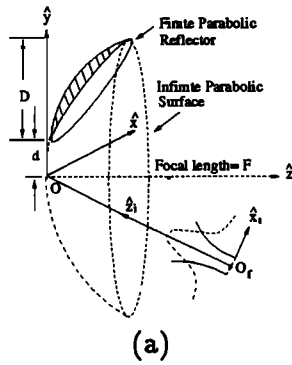


Figure 6. (a) Geometry of the 3-D Gaussian beam scattering from a finite curved surface. (b) Beam reflection point is on the edge, $\phi = 0$ degrees plane. (c) Beam waist located at $(0, -4\lambda, 36\lambda)$, $\phi = 90$ degrees plane. $F = 36\lambda$, $D = 40\lambda$, and offset, $d = 2\lambda$.

Figure 7. (a) Geometry of the 3-D Gaussian Beam scattering from a finite curved surface. (b) Beam reflection point is on the surface, $\phi = 0$ degrees plane. (c) Beam waist is located at $(0, -36\lambda, 36\lambda)$, $\phi = 90$ degrees plane. $F = 36\lambda$, $D = 40\lambda$, and $d = 4\lambda$.

seen from Figures 5–7 that these solutions based on the two independent procedures are found to agree extremely well. Furthermore, it is found that the GB based approach developed here is orders of magnitude faster than the numerical PO integration approach when dealing with realistic feed illumination which is then expanded in a set of GBs (Chou et al., submitted manuscript, 1996). In each of the cases in Figures 5–7, the incident GB is taken to be rotationally symmetric with the associated incident magnetic field $\bar{H}^i$ to be given by

$$\bar{H}^i(x_i, y_i, z_i) = \hat{y}_i \frac{\rho + jb}{\rho + jb + z_i} e^{-jk \left[z_i + \frac{1}{2} \frac{x_i^2 + y_i^2}{\rho + jb + z_i} \right]}. \quad (76)$$

The specific values of the complex radius of curvature $\rho + jb$ are indicated in Figures 5–7. In particular, $b = 12.53\lambda$ and ρ is the distance from O_f , the waist of the incident GB, to point O . It is noted that ρ is different for each of the three cases in Figures 5, 6, and 7, respectively.

Appendix A: Expressions Required to Compute the Jacobian of the Transformation From (x, y) to (ρ, ϕ)

$$\frac{\partial x}{\partial \rho_1} = A_1 \cos \phi_1 + A_2 \sin \phi_1 \quad (A1a)$$

$$\frac{\partial x}{\partial \phi_1} = [-A_1 \sin \phi_1 + A_2 \cos \phi_1] \rho_1 \quad (A1b)$$

$$\frac{\partial y}{\partial \rho_1} = B_1 \cos \phi_1 + B_2 \sin \phi_1 \quad (A2a)$$

$$\frac{\partial y}{\partial \phi_1} = [-B_1 \sin \phi_1 + B_2 \cos \phi_1] \rho_1 \quad (A2b)$$

The second derivatives can be found by

$$\frac{\partial^2 x}{\partial \rho_1^2} = 0 \quad (A3a)$$

$$\frac{\partial^2 x}{\partial \phi_1^2} = -[A_1 \cos \phi_1 + A_2 \sin \phi_1] \rho_1 \quad (A3b)$$

$$\frac{\partial^2 y}{\partial \rho_1^2} = 0 \quad (A4a)$$

$$\frac{\partial^2 y}{\partial \phi_1^2} = -[B_1 \cos \phi_1 + B_2 \sin \phi_1] \rho_1. \quad (A4b)$$

Appendix B: Evaluation of the Parameter ϕ_{1se} in the GB Edge Diffraction Formula

The value of ϕ_{1se} defined in (70) and encountered in (72) is found here as follows. From (61)

$$\begin{aligned} g_\rho(\rho_1 = d) &= f(x_{se}, y_{se}) \\ &= f(x_s, y_s) - a_0(x_{se} - x_s)^2 - \\ &\quad a_1(y_{se} - y_s)^2 - 2c(x_{se} - x_s) \cdot \\ &\quad (y_{se} - y_s) \end{aligned} \quad (B1)$$

where $x = x_{se}$ and $y = y_{se}$ when $\rho_1 = d$. Also,

$$\left. \frac{\partial g_\rho}{\partial \rho_1} \right|_{\rho_1=d} = g'_\rho(d) \quad (B2)$$

where

$$\begin{aligned} g'_\rho(d) &= [-2a_0(x_{se} - x_s) - 2c(y_{se} - y_s)] \cdot \\ &\quad (A \cos \psi_{1se}) + [-2a_1(y_{se} - y_s) - \\ &\quad 2c(x_{se} - x_s)] \cdot (B \sin \psi_{1se}) \end{aligned} \quad (B3)$$

in which ψ_{1se} is obtained from (47) as

$$\psi_{1se} = \phi_{1se} + \phi_0. \quad (B4)$$

It can be shown that

$$\begin{aligned} g''_\phi(\phi_{1se}) \Big|_{\rho_1=d} &= -dg'_\rho(d) - 2d^2 \left[a_0 A^2 \cdot \right. \\ &\quad \sin^2 \psi_{1se} - 2cAB \sin \psi_{1se} \cdot \\ &\quad \left. \cos \psi_{1se} + a_1 B^2 \cdot \cos^2 \psi_{1se} \right]. \end{aligned} \quad (B5)$$

In order to evaluate $g'_\rho(d)$ and $g''_\phi(\phi_{1se}) \Big|_{\rho_1=d}$ of (B3) and (B5) which also appear in the expression for $\bar{P}_d(\bar{r})$ of (72), it is important to find

ϕ_{1se} which, in turn, depends on ϕ_{1se} . The ϕ_{1se} is found from the condition

$$g'_\phi(\phi_{1se}) \Big|_{\rho_1=d} = 0 \quad (\text{B6})$$

which can be written explicitly as

$$H_1 \sin^2 \phi_{1se} + H_2 \sin \phi_{1se} \cos \phi_{1se} + H_3 \sin \phi_{1se} H_4 \cos \phi_{1se} + H_5 = 0 \quad (\text{B7})$$

where

$$\begin{aligned} H_1 &\equiv 4d[a_0 A_1 A_2 + a_1 B_1 B_2 + c(A_2 B_1 + B_2 A_1)] \\ &= 2d[(-a_0 A^2 + a_1 B^2) \sin 2\phi_0 + 2cAB \cos 2\phi_0] \end{aligned} \quad (\text{B8})$$

$$\begin{aligned} H_2 &\equiv 2d[a_0(A_1^2 - A_2^2) + a_1(B_1^2 - B_2^2) + 2c(A_1 B_1 - A_2 B_2)] \\ &= 2d[(a_0 A^2 - a_1 B^2) \cos 2\phi_0 + 2cAB \sin 2\phi_0] \end{aligned} \quad (\text{B9})$$

$$H_3 \equiv 2\{a_0 A_1(x_o - x_s) + a_1 B_1(y_o - y_s) + c[A_1(y_o - y_s) + B_1(x_o - x_s)]\} \quad (\text{B10})$$

$$H_4 \equiv -2\{a_0 A_2(x_o - x_s) + a_1 B_2(y_o - y_s) + c[A_2(y_o - y_s) + B_2(x_o - x_s)]\} \quad (\text{B11})$$

$$H_5 \equiv -\frac{1}{2}H_1. \quad (\text{B12})$$

It can be demonstrated that ϕ_{1se} in (B7) can be found in closed form by replacing $\cos \phi_{1se}$ with $\sqrt{1 - \sin^2 \phi_{1se}}$ and then solving a quartic equation for $\sin \phi_{1se}$ at the expense of some tedious algebra. However, a significant simplification in the algebra is possible with negligible error if one approximates $\sqrt{1 - \sin^2 \phi_{1se}}$ for $\cos \phi_{1se}$ as

$$\cos \phi_{1se} \approx 1 - \frac{\sin^2 \phi_{1se}}{2}. \quad (\text{B13})$$

That could be taken to mean that $\sin^2 \phi_{1se} \ll 1$, which is generally true for $\phi_{1se} < 15^\circ$. Thus one might include the criteria that $\sin \phi_{1se} < 1/4$ for practical purposes. The above simplification valid for small ϕ_{1se} requires that the incident GB spot size be small compared to the surface of the reflector. The latter condition is easily met for practical reflector antennas, which are electrically large. Otherwise, one must resort to the more tedious algebra. Incorporating (B13) into (B7) yields a cubic equation (instead of a quartic) in $\sin \phi_{1se}$ as given by

$$\begin{aligned} \sin^3 \phi_{1se} + \frac{(H_4 - 2H_1)}{H_2} \sin^2 \phi_{1se} - \\ \frac{2(H_2 + H_3)}{H_2} \sin \phi_{1se} - \frac{2(H_4 + H_5)}{H_2} = 0 \end{aligned} \quad (\text{B14})$$

which is of the general form

$$z^3 + \alpha_1 z^2 + \alpha_2 z + \alpha_3 = 0 \quad (\text{B15})$$

with roots $z = z_1$, $z = z_2$ and $z = z_3$ given by

$$z_1 = S + T - \frac{1}{3}\alpha_1 \quad (\text{B16})$$

$$z_2 = -\frac{1}{2}(S + T) - \frac{1}{3}\alpha_1 + \frac{1}{2}j\sqrt{3}(S - T) \quad (\text{B17})$$

$$z_3 = -\frac{1}{2}(S + T) - \frac{1}{3}\alpha_1 - \frac{1}{2}j\sqrt{3}(S + T) \quad (\text{B18})$$

in which

$$S = \sqrt[3]{R + \sqrt{Q^3 + R^2}} \quad (\text{B19})$$

$$T = \sqrt[3]{R - \sqrt{Q^3 + R^2}} \quad (\text{B20})$$

$$Q = \frac{3\alpha_2 - \alpha_1^2}{9} \quad (\text{B21})$$

$$R = \frac{9\alpha_1\alpha_2 - 27\alpha_3 - 2\alpha_1^3}{54} \quad (\text{B22})$$

with the condition that

$$S \times T = -Q \quad (\text{B23})$$

since the coefficients α_1 , α_2 and α_3 are in general complex valued.

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